

Sardar Patel Institute of Technology
(Electronics & Telecommunication Department)



RS Lab

Experiment 1

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PROBLEM STATEMENT:

$U = A \cdot \sin(\theta) \cdot \cos(n \cdot \theta)$ where n is your last digit UCID no. if $n=0$ then consider $n+1$

AIM: To calculate and infer various fundamental parameters of antenna like radiation pattern, Radiation power density, Radiation Intensity, Gain, Directivity, HPBW and FNBW using Scilab/Matlab.

OBJECTIVE: calculate, plot, and analyze the fundamental parameters of an antenna such as radiation pattern, radiation power density, radiation intensity, directivity, gain, half power beam width (HPBW), and first null beam width (FNBW) using MATLAB/SCILAB, and to infer the radiation characteristics and performance of the antenna.

INPUT SPECIFICATIONS:

Amplitude: $A=1$

Angle range: $0 \leq \theta \leq 2\pi$

$n=5$

PROBLEM DEFINITION:

- 1) To plot the Radiation Pattern.
- 2) To compute HPBW.
- 3) To compute FNBW.
- 4) To compute Directivity.
- 5) To compute Gain
- 6) Write the inference and tabulate the results for various performance parameters of Antenna.
- 7) Add handwritten solution

THEORY :

An antenna radiation pattern describes how the radiated power of an antenna is distributed in space as a function of angle.

In this experiment, the radiation pattern depends on the angular variation of the field. The value of $n = 5$ introduces multiple oscillations in the pattern, resulting in a multi-lobed radiation structure. Because of this, the antenna does not radiate uniformly but forms a main lobe along with several side lobes and nulls.

Radiation Intensity

Radiation intensity represents the power radiated by the antenna in a particular direction per unit solid angle. It determines the shape and strength of the radiation pattern.

Radiation Power Density

Radiation power density is the amount of power passing through a unit area at a certain distance from the antenna. It decreases as the distance from the antenna increases and follows the inverse square law.

Total Radiated Power

Total radiated power is the overall power emitted by the antenna in all directions. It is obtained by considering radiation over the complete angular range.

Directivity

Directivity indicates how focused the antenna radiation is in a particular direction. An antenna with higher directivity concentrates more power in its main lobe compared to other directions. Since $n = 5$, the pattern becomes more directional due to multiple oscillations, which generally increases directivity.

Gain

Gain represents the practical performance of the antenna and includes the effect of antenna efficiency. For an ideal antenna (100% efficient), gain is equal to directivity.

Half Power Beam Width (HPBW)

HPBW is the angular width of the main lobe measured between the two points where the power reduces to half of its maximum value. A smaller HPBW indicates better focusing capability.

First Null Beam Width (FNBW)

FNBW is the angular separation between the first null points on either side of the main lobe. It shows the total spread of the main radiation region.

CODE :

```
% Fundamental Parameters of Antenna
%Rushikesh Bathe -2023200015
clc;
clear;
close all;
n = 5;
A = 1;
theta = linspace(0, pi, 1000);
% Radiation Intensity
U = A .* sin(theta) .* cos(n * theta);
% Power Pattern
P = abs(U).^2;
% Normalized Power
P_norm = P ./ max(P);
% Radiation Pattern Plot
figure;
polarplot(theta, P_norm, 'LineWidth', 2);
title('Radiation Pattern for U(\theta) = sin(\theta)cos(5\theta)');
grid on;
% HPBW Calculation (-3 dB points)
P_dB = 10 * log10(P_norm);
idx_hpbw = find(P_dB >= -3);
HPBW = (theta(idx_hpbw(end)) - theta(idx_hpbw(1))) * (180/pi);
% FNBW Calculation (First Nulls)
null_idx = find(abs(U) < 1e-3);
FNBW = (theta(null_idx(end)) - theta(null_idx(1))) * (180/pi);
% Directivity Calculation
Umax = max(P);
Uavg = mean(P);
Directivity = Umax / Uavg;
% Gain Calculation
```

```
eta = 0.8;
Gain = eta * Directivity;
% Display Results
disp('-----');
disp('Fundamental Antenna Parameters');
disp('-----');
fprintf('HPBW      = %.4f degrees\n', HPBW);
fprintf('FNBW      = %.4f degrees\n', FNBW);
fprintf('Directivity = %.4f\n', Directivity);
fprintf('Gain       = %.4f\n', Gain);
disp('-----');
```

OUTPUT :

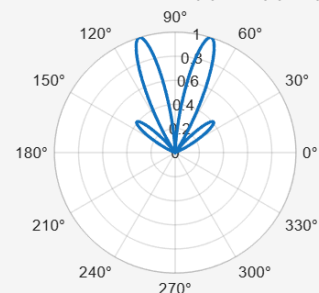
Command Window

New to MATLAB? See resources for [Getting Started](#).

Fundamental Antenna Parameters

```
HPBW      = 52.4324 degrees
FNBW      = 180.0000 degrees
Directivity = 3.6361
Gain       = 2.9089
```

Radiation Pattern for $U(\theta) = \sin(\theta)\cos(5\theta)$



HANDWRITTEN SOLUTION

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Radiating System

Exp 1

Aim:- To calculate & infer various fundamental parameters of antenna like radiation pattern, Radiation power density, Radiation intensity, gain, Directivity, HPBW & FNBW using Matlab.

Problem definition:

- 1) To plot the Radiation pattern
- 2) To compute HPBW
- 3) To compute FNBW
- 4) To compute Directivity
- 5) To compute Directivity gain
- 6) write the interference & calculate the result for various performance parameters of Antenna
- 7) Add handwritten soln

Calculation

$$U(\theta) = A \sin \theta \cos \theta$$

$$n=5, A=1$$

$$U(\theta) = \sin \theta \cos \theta$$

1) Radiation Power Pattern

$$P = \frac{1}{2} |U(\theta)|^2$$

$$P(\theta) = \sin^2(\theta) \cos^2(\theta)$$

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Max radiation intensity occurs when
 $\sin \theta \cos \theta = 1$
 $P_{max} = 1$

2) Half Power Beamwidth (HPBW)
 ↳ Angular separⁿ b/w points where power drops to half of maximum power
 $P(\theta) = \frac{1}{2} P_{max}$

$$\sin^2 \theta \cos^2 \theta = 0.5$$

$$\theta_1 = 63.78^\circ, \theta_2 = 116.21^\circ$$

$$HPBW = \theta_2 - \theta_1$$

$$= 116.21^\circ - 63.78^\circ$$

$$HPBW = 52.4324^\circ$$

3) First Null Beamwidth (FNBW)

Null condⁿ $\rightarrow U(\theta) = 0$
 occurs when $\sin \theta = 0$ or $\cos \theta = 0$
 First null occur at $\theta = 0^\circ$ & $\theta = 180^\circ$
 $FNBW = 180^\circ$

4) Directivity (D) Avg Radiation Intensity U_{avg}
 $U_{avg} = \frac{P_{rad}}{4\pi}$

Total radiated power: $P_{rad} = \int_0^{2\pi} \int_0^\pi U(\theta) \sin \theta d\theta d\phi$

$$V_{avg} = \frac{1}{4\pi} \int_0^{\pi} \int_0^{2\pi} U(\theta) \sin\theta d\theta d\phi$$

$$V_{avg} = \frac{1}{4\pi} \int_0^{\pi} \sin^2\theta \cos^2\theta d\theta \int_0^{2\pi} d\phi$$

$$= \frac{1}{4} \int_0^{\pi} \sin^2\theta (1 + \cos 10\theta) d\theta$$

$$V_{avg} = \frac{1}{4} \left[\int_0^{\pi} \sin^2\theta d\theta + \int_0^{\pi} \sin^2\theta \cos 10\theta d\theta \right]$$

$$V_{avg} = \frac{\pi^2}{8} \approx 0.3927$$

4) Directivity

$$D = \frac{U_{max}}{V_{avg}}$$

$$= \frac{1}{0.3927} = 3.6361$$

5) Gain

$$G = \eta D$$

$$G = 0.8 \times 3.6361$$

$$= 2.9089$$

S.No	Parameter	Symbol	Value
1	Radiation Intensity	$U(\theta)$	$\sin\theta \cos^2\theta$
2	Max. Radiation Intensity	U_{max}	1
3	Average Rad ⁿ Inten ^y	U_{avg}	0.3927
4	Half Power BW	HPBW	52.43°
5	First Null Beamwidth	FNBW	180°
6	Directivity	D	3.6361
7	Gain	G	2.9089

Intereracy

The radiation pattern for the antenna for $n=5$ shows a directive behaviour with maximum radiation occurring at broadside ($\theta=90^\circ$). The half power beamwidth is moderate, indicating good angular resolution. The calculated directivity and gain values confirm that the antenna effectively concentrates radiated power in the main lobe. As the value of n increases, the antenna becomes more directional with reduced beamwidth, improving overall performance.

CONCLUSION :

The radiation pattern of the antenna for $n=5$ shows a directive behavior with maximum radiation occurring at broadside ($\theta=90^\circ$). The half power beamwidth is moderate, indicating good angular resolution. The calculated directivity and gain values confirm that the antenna effectively concentrates radiated power in the main lobe. As the value of n increases, the antenna becomes more directional with reduced beamwidth, improving overall performance.